

- 15** A mutation occurred within the DNA sequence coding for an enzyme, causing a decrease in the rate of a reaction catalysed by this mutated enzyme.

Which statements could explain the decrease in the rate of reaction?

- 1 An inhibitor for this enzyme has an increased affinity for the enzyme and forms an enzyme–inhibitor complex more easily.
- 2 The active site of the enzyme might have changed shape and so is less complementary.
- 3 The activation energy for the reaction with the mutated enzyme is greater than with the non-mutated enzyme.

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only